

Universidad Rafael Landívar

Facultad de Ingeniería

Ingeniería en Informática y Sistemas

Redes y Telecomunicaciones- Sección 1

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LABORATORIO 13

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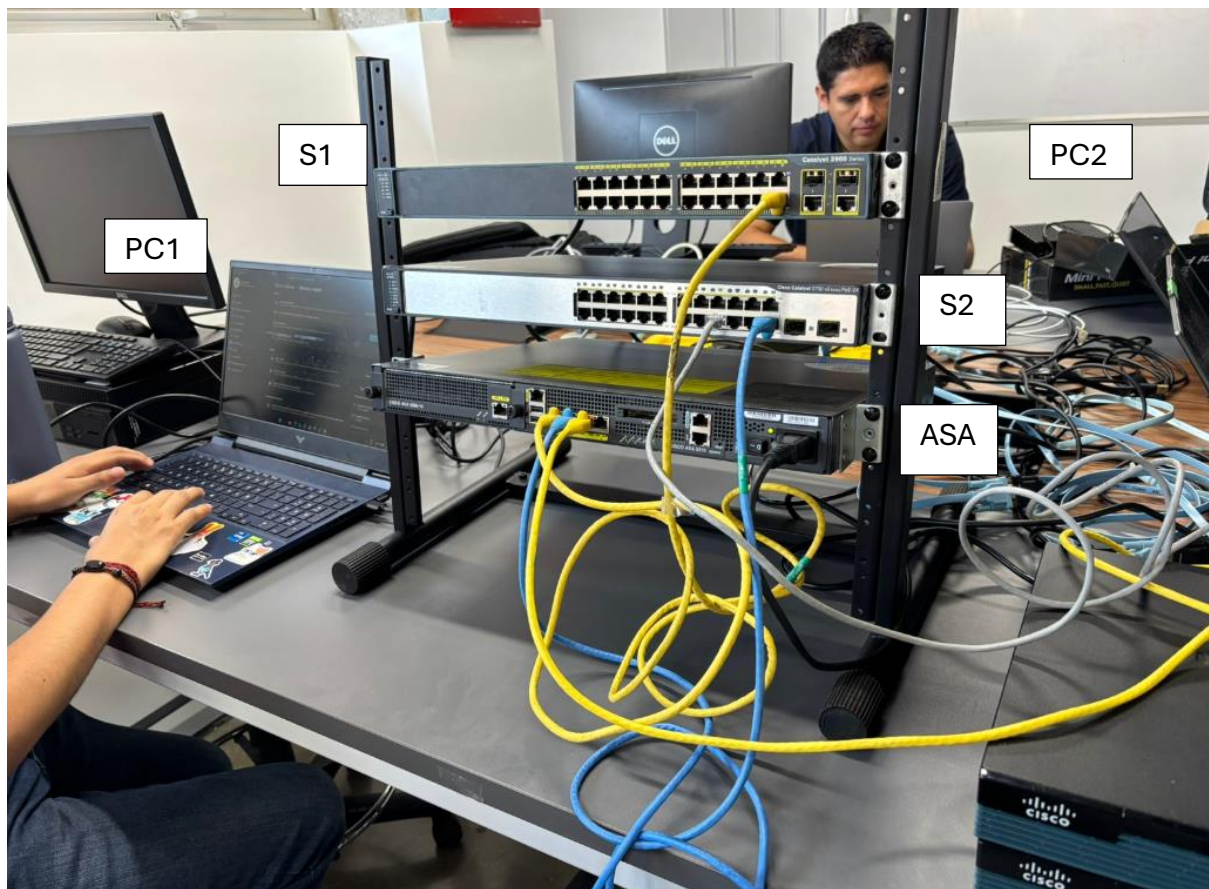
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Viernes, 10 de abril de 2026

Capturas de topología de red



```

R2#
R2#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down  down
GigabitEthernet0/0   10.1.1.2        YES manual  down        down
GigabitEthernet0/1   10.2.2.2        YES manual  down        down
Loopback0            172.30.1.1      YES manual  up          up
R2#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down  down
GigabitEthernet0/0   10.1.1.2        YES manual  down        down
GigabitEthernet0/1   10.2.2.2        YES manual  down        down
Loopback0            172.30.1.1      YES manual  up          up
R2#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down  down
GigabitEthernet0/0   10.1.1.2        YES manual  down        down
GigabitEthernet0/1   10.2.2.2        YES manual  down        down
Loopback0            172.30.1.1      YES manual  up          up
R2#
May  2 15:03:18.227: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
May  2 15:03:19.227: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R2#
R2#
R2#
R2#ping 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
R2#ping 209.165.200.227
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.227, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R2#ping 209.165.200.227
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.227, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R2#ping 192.168.1.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.3, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R2#

```

```

R1#ping 209.165.200.226
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.226, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1#ping 10.1.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R1#ping 172.30.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.30.1.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
R1#

```

```

Password:
Password:
S2>en
Password:
S2#
S2#
S2#show interface trunk          ! Fa1/0/24 trunking VLANs 10,20

Port      Mode      Encapsulation  Status      Native vlan
Fa1/0/24   on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa1/0/24   10,20

Port      Vlans allowed and active in management domain
Fa1/0/24   10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa1/0/24   10,20
S2#show vlan brief              ! VLANs 10 and 20 active

VLAN Name                Status    Ports
-----
1    default                active    Fa1/0/1, Fa1/0/2, Fa1/0/3
                                           Fa1/0/4, Fa1/0/5, Fa1/0/6
                                           Fa1/0/7, Fa1/0/8, Fa1/0/9
                                           Fa1/0/10, Fa1/0/11, Fa1/0/12
                                           Fa1/0/13, Fa1/0/14, Fa1/0/15
                                           Fa1/0/16, Fa1/0/17, Fa1/0/19
                                           Fa1/0/20, Fa1/0/21, Fa1/0/22
                                           Fa1/0/23, Gi1/0/2
10   DMZ                     active
20   INSIDE                  active    Fa1/0/18
30   Vlan30                  active
100  Vlan100                  active
1002 fddi-default           act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default       act/unsup
1005 trnet-default         act/unsup
S2#show ip interface brief    ! Vlan10 and Vlan20 up/up
Interface      IP-Address  OK? Method Status      Protocol
Vlan1          unassigned  YES manual administratively down down
Vlan10         192.168.2.2 YES manual up          up
Vlan20         unassigned  YES unset  up          up
FastEthernet1/0/1 unassigned YES unset  down       down
FastEthernet1/0/2 unassigned YES unset  down       down
FastEthernet1/0/3 unassigned YES unset  down       down
FastEthernet1/0/4 unassigned YES unset  down       down
FastEthernet1/0/5 unassigned YES unset  down       down
FastEthernet1/0/6 unassigned YES unset  down       down
FastEthernet1/0/7 unassigned YES unset  down       down
FastEthernet1/0/8 unassigned YES unset  down       down
FastEthernet1/0/9 unassigned YES unset  down       down
FastEthernet1/0/10 unassigned YES unset  down       down
FastEthernet1/0/11 unassigned YES unset  down       down

```

```

10   DMZ                active
20   INSIDE             active   Fa1/0/18
30   Vlan30             active
100  Vlan100            active
1002 fddi-default       act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default    act/unsup
1005 trnet-default      act/unsup
S2#show ip interface brief      ! Vlan10 and Vlan20 up/up
Interface      IP-Address      OK? Method Status      Protocol
Vlan1          unassigned      YES manual administratively down down
Vlan10         192.168.2.2     YES manual up          up
Vlan20         unassigned      YES unset  up          up
FastEthernet1/0/1 unassigned      YES unset  down        down
FastEthernet1/0/2 unassigned      YES unset  down        down
FastEthernet1/0/3 unassigned      YES unset  down        down
FastEthernet1/0/4 unassigned      YES unset  down        down
FastEthernet1/0/5 unassigned      YES unset  down        down
FastEthernet1/0/6 unassigned      YES unset  down        down
FastEthernet1/0/7 unassigned      YES unset  down        down
FastEthernet1/0/8 unassigned      YES unset  down        down
FastEthernet1/0/9 unassigned      YES unset  down        down
FastEthernet1/0/10 unassigned      YES unset  down        down
FastEthernet1/0/11 unassigned      YES unset  down        down
FastEthernet1/0/12 unassigned      YES unset  down        down
FastEthernet1/0/13 unassigned      YES unset  down        down
FastEthernet1/0/14 unassigned      YES unset  down        down
FastEthernet1/0/15 unassigned      YES unset  down        down
FastEthernet1/0/16 unassigned      YES unset  down        down
FastEthernet1/0/17 unassigned      YES unset  down        down
FastEthernet1/0/18 unassigned      YES unset  up          up
FastEthernet1/0/19 unassigned      YES unset  down        down

S2#show ip route                ! Routing table
^
% Invalid input detected at '^' marker.

S2#show ip route                ! Routing table
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.2.0/24 is directly connected, Vlan10
S2#
S2#

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En el laboratorio se logro la conectividad entre swiches según la topología aplicada obviando la parte del rauter R3 y el swich S3 en la configuración por los materiales disponibles en el laboratorio, en este caso se aplico algunas ip para verificar la conectividad de todos los componentes y hacer las vlans para lograr hacer el trafico de datos y los pings para revisar el trafico entre los componentes.